

The University of Lahore

Department of Computer Science & IT

Final-term Examination

Fall -2022

Course Title:	Advanced Software Engineering	Course Code:	CS4327	Credit Hours:	3
Course Instructor:	Mr. Sajid Ullah Khan, Mr. Waqar Ashiq, Mr. Ubaid Azam, Ms. Ushna Khalil	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum	40		
Date:	12-01-2023	Course Mentor:	Dr. Adnan Khalid		
Student's Name:		Signature:			
Reg. No:		Section:			

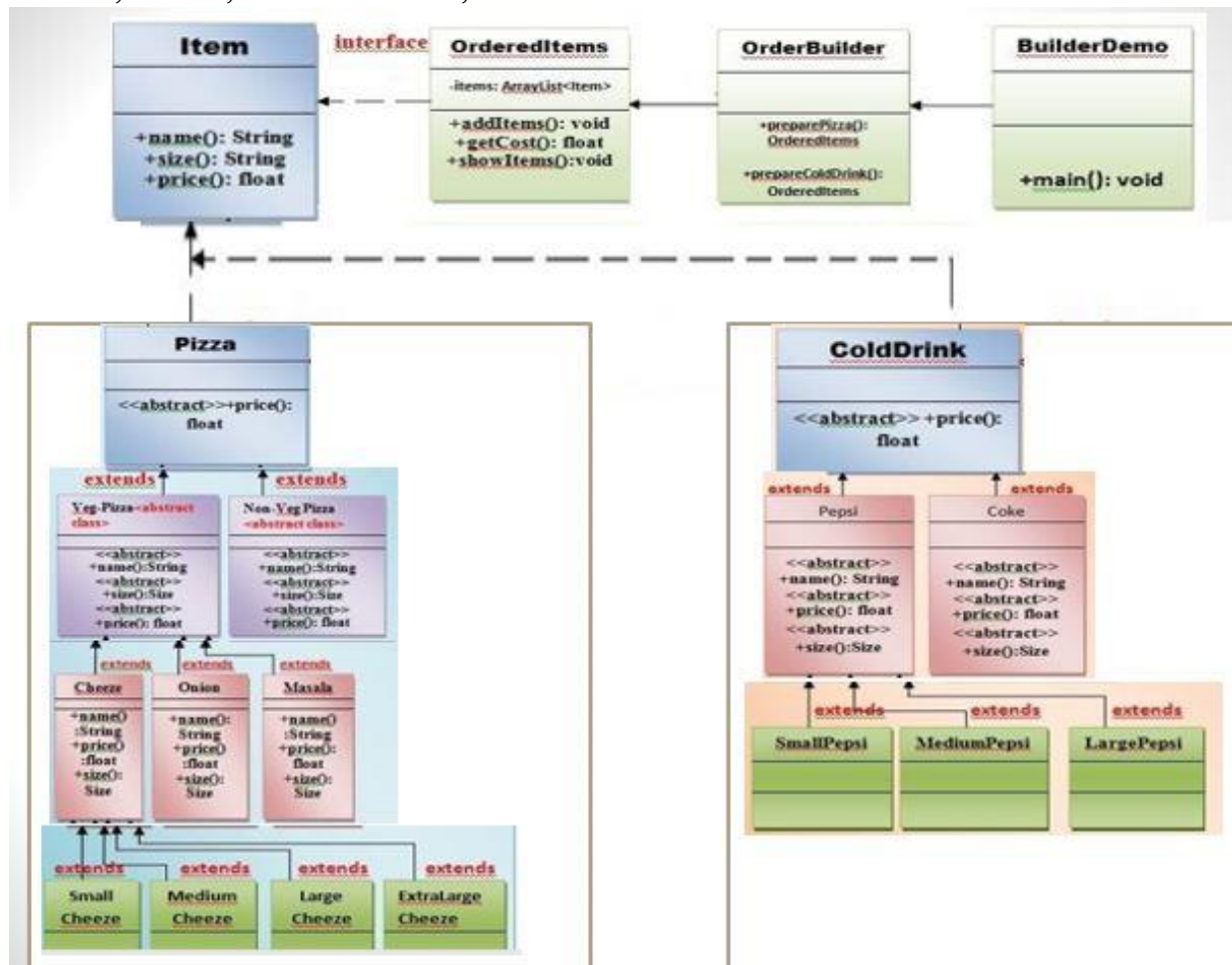
Question #1:

10 Marks

- a) Explain elements of builder Pattern with the help of any case study? You may attempt a Pizza hut case study as discussed in class.

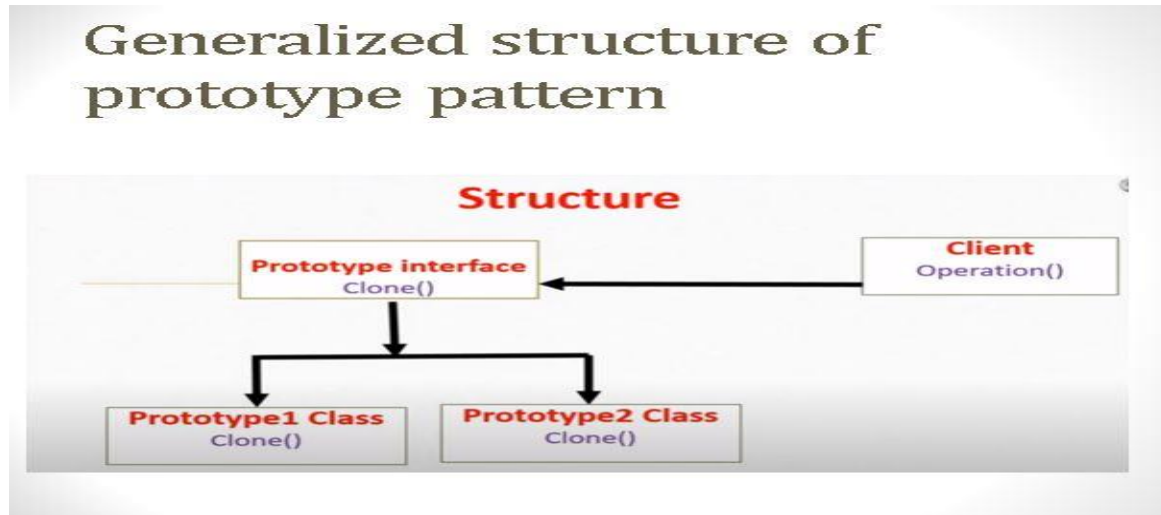
Sol: Following are the elements of builder pattern:

Director, Builder, Concrete Builder, Product



b) Draw a generalized structure of a Prototype pattern?

Sol:



Question #2:

10 Marks

Consider a scenario in which you are hired by a software house that's currently at level 2 of CMM. The CEO wants you to take the necessary actions so that their software house can achieve level 3 of CMM. What necessary steps you will take, explain them in detail.

ANS: To achieve level 3 from level 2, I will implement following Key process Areas (KPA) in that software house.

- Organization Process Focus

Establish organizational responsibility for software process activities that improve the organization's overall software process capability

- Organization Process Definition

Develop and maintain a usable set of software process assets stable foundation that can be established

basis for defining meaningful data for quantitative process management

- Training Program

Develop skills and knowledge so that individual can perform their roles effectively and efficiently

Organizational responsibility

Needs identified by project

- Integrated Software Management

Integrated engineering and management activities

Engineering and management processes are custom-made from the organizational standard processes

Modifying based on business environment and project needs

- Software Product Engineering

technical activities of the project are well defined (SDLC)

correct, consistent work products

- Intergroup Coordination

Software engineering groups participate actively with other groups

- Peer Reviews

early defect detection and removal

better understanding of the products

implemented with inspections, walkthroughs, etc

Question #3:

10 Marks

a) Differentiate Software Testing and Software Quality Assurance.

SOL: Testing focuses on system control and error detection, with product orientation and corrective actions. Testing is about checking the behavior of the application, while quality assurance is about making the overall quality level of the project better each day.

b) Being a software developer when you will use Test Driven Development (TDD)? Explain its diagram, process activities, and benefits in detail.

SOL:

: Test-driven development (TDD) is an approach to program development in which you inter-leave testing and code development. You develop code incrementally, along with a test for that increment. You don't move on to the next increment until the code that you have developed passes its test.

Process Activities:

Start by identifying the increment of functionality that is required. This should normally be small and implementable in a few lines of code. Write a test for this functionality and implement this as an automated test.

Run the test, along with all other tests that have been implemented. Initially, you have not implemented the functionality so the new test will fail. Implement the functionality and re-run the test. Once all tests run successfully, you move on to implementing the next chunk of functionality.

Benefits:

- Code coverage

Every code segment that you write has at least one associated test so all code written has at least one test.

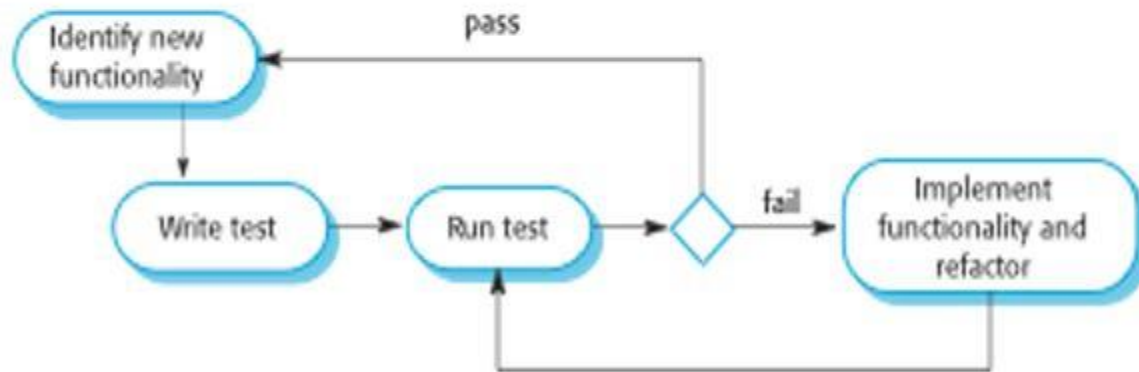
- Regression testing

A regression test suite is developed incrementally as a program is developed.

- Simplified debugging

When a test fails, it should be obvious where the problem lies. The newly written code needs to be checked and modified.

Test-driven development



Question #4:**10 Marks**

- a) Describe the categories of software maintenance in brief and to the point? What are the common priorities to change requests?

Sol: Following are the categories of software maintenance:

Corrective maintenance

Adaptive maintenance

Perfective maintenance

Preventive maintenance

Priorities: Urgent, Critical, Major, Minor

- b) Why maintenance costs are too high? Explain with the help of a scenario.

Sol:

We cannot trace the product nor the process that created the product.

Changes are not adequately documented.

The notion that software maintenance is essentially a post-delivery activity.

The user compares software problems to hardware problems; if bugs occur or parts break, fix them